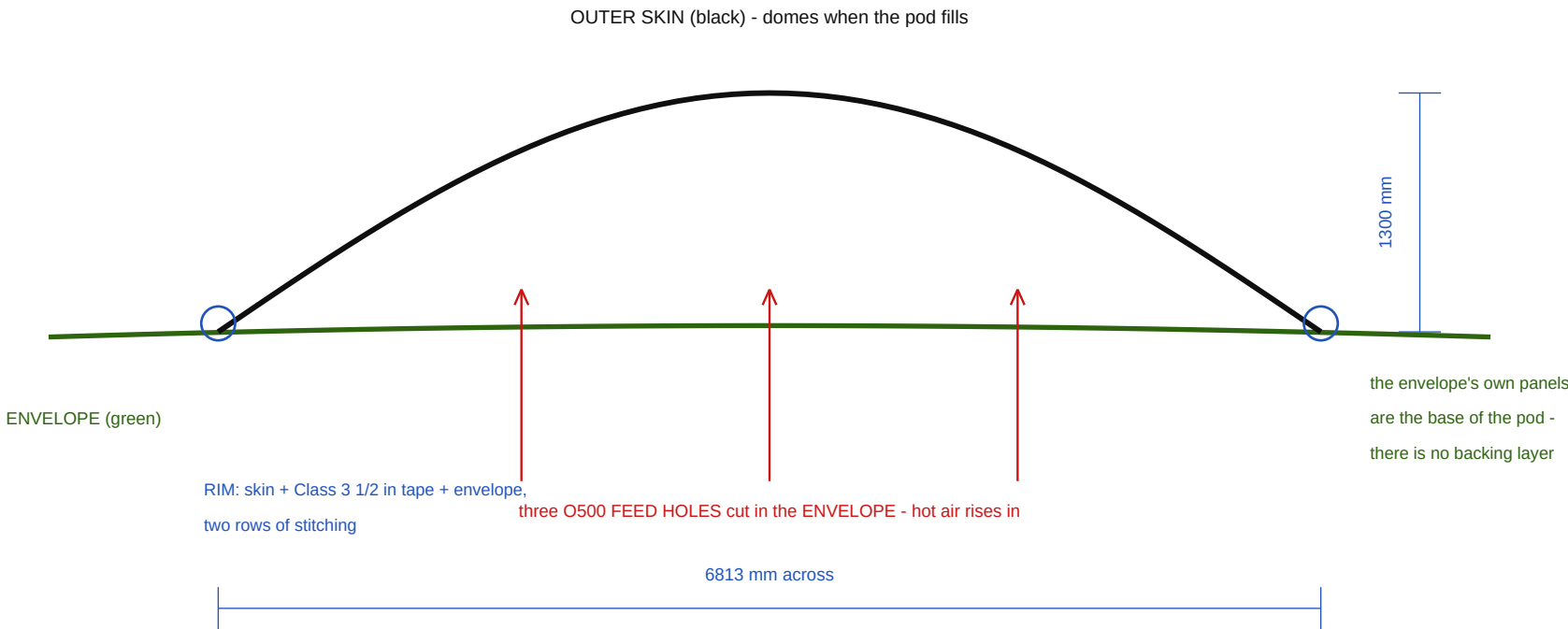


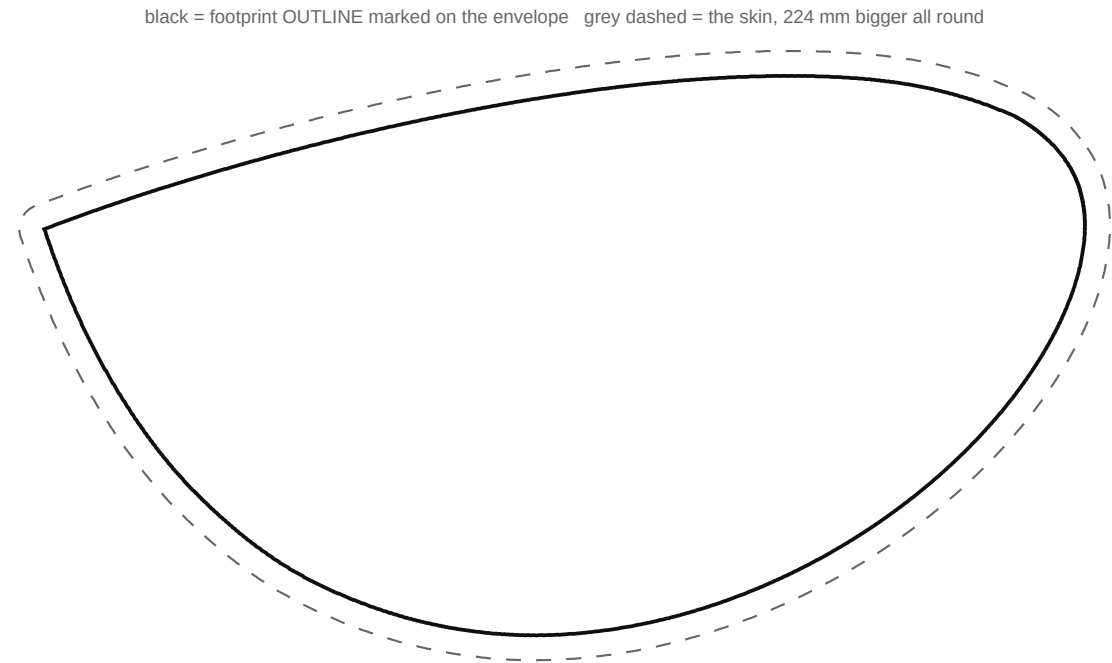
EYE POD - WHAT YOU ARE BUILDING

a ram-air blister: ONE black skin sewn straight to the envelope's own gore panels over Class 3 1/2 in tape, inflated through three holes cut in the envelope

SECTION THROUGH THE POD (across the short axis)



PLAN - where it sits on the envelope



LEFT pod: gores 18-20. RIGHT pod: gores 2-4, a MIRROR of every piece.

Rows F to J - the pod spans three gores and five panel rows exactly.

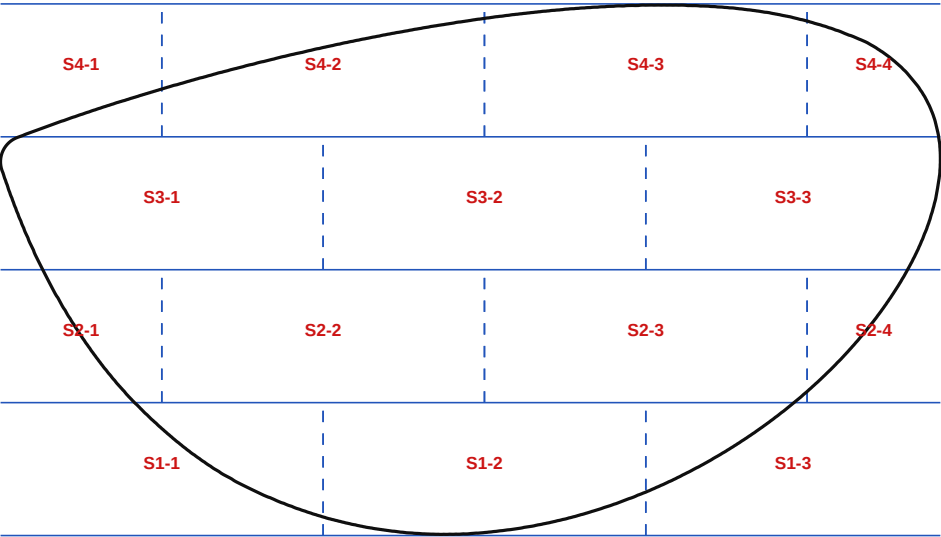
Feed holes: gores 20/2 panel H +450 mm; gores 19/3 panel H +910 mm; gores 18/4 panel i +850 mm

Layers	envelope gore panels + Class 3 1/2 in (12.7 mm) load tape + black skin. There is no backing layer.
Footprint	7.96 x 6.81 m, rim 23.18 m - MARKED on the envelope, not cut
Outer skin	footprint offset OUTWARD 224 mm - 115.5 % of the footprint area
Dome height	1300 mm proud at the apex when inflated
Rim ease	1.24 m (5.4 %) - 78 mm per segment over 16 match points
Pieces per pod	14 skin pieces - none over 1450 x 3450 mm, so every one prints 1:1
Cloth	~37 running m of 60 in per pod, ~73 m for both

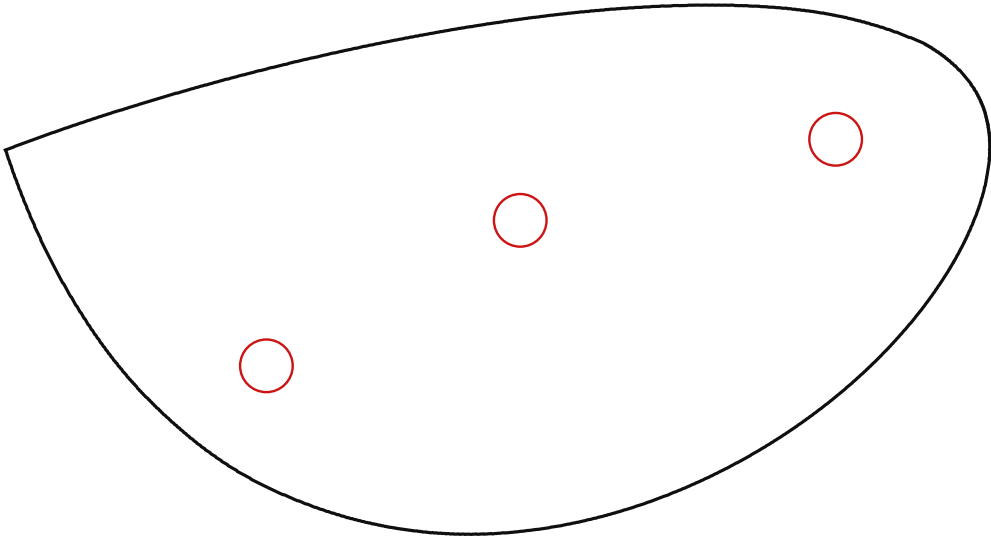
THE PIECES

the skin is 9903 mm across at cut size and cannot be cut whole from 60 in cloth - it is a jigsaw of 14 pieces, and every one prints 1:1 in alien-balloon-eye-patterns.pdf

OUTER SKIN - the only cut layer



FOOTPRINT - marked on the envelope



no cloth is cut to this outline; the red circles are the feed holes, cut in the envelope panels
Strip seams (solid blue) are dead straight; section joints (dashed) are staggered so no four seams meet at a point. The skin carries no holes - the feed holes are cut in the envelope, clear of the rim by 700 mm or more.

CUT LIST - every size INCLUDES the 25 mm allowance already added on the seam edges

piece	layer	cut w x h mm	pattern
S1-1	SKIN	2012 x 1229	one curved edge
S1-2	SKIN	3450 x 1413	one curved edge
S1-3	SKIN	1586 x 968	one curved edge
S2-1	SKIN	1279 x 1450	one curved edge
S2-2	SKIN	3450 x 1450	rectangle
S2-3	SKIN	3450 x 1450	one curved edge
S2-4	SKIN	1083 x 1333	one curved edge

piece	layer	cut w x h mm	pattern
S3-1	SKIN	3424 x 1450	one curved edge
S3-2	SKIN	3450 x 1450	rectangle
S3-3	SKIN	3129 x 1450	one curved edge
S4-1	SKIN	1528 x 529	one curved edge
S4-2	SKIN	3450 x 1272	one curved edge
S4-3	SKIN	3450 x 1413	one curved edge
S4-4	SKIN	1411 x 1244	one curved edge

Every piece above is cut TWICE: once as printed for the LEFT pod, once mirrored for the RIGHT.

That is 28 pieces of black ripstop in total, about 73 running metres of 60 in cloth - the envelope supplies the other side of the pod.

STEPS 1-2 - CUT AND LABEL

get this right and the rest is assembly; get it wrong and you will not find out until the rim will not close

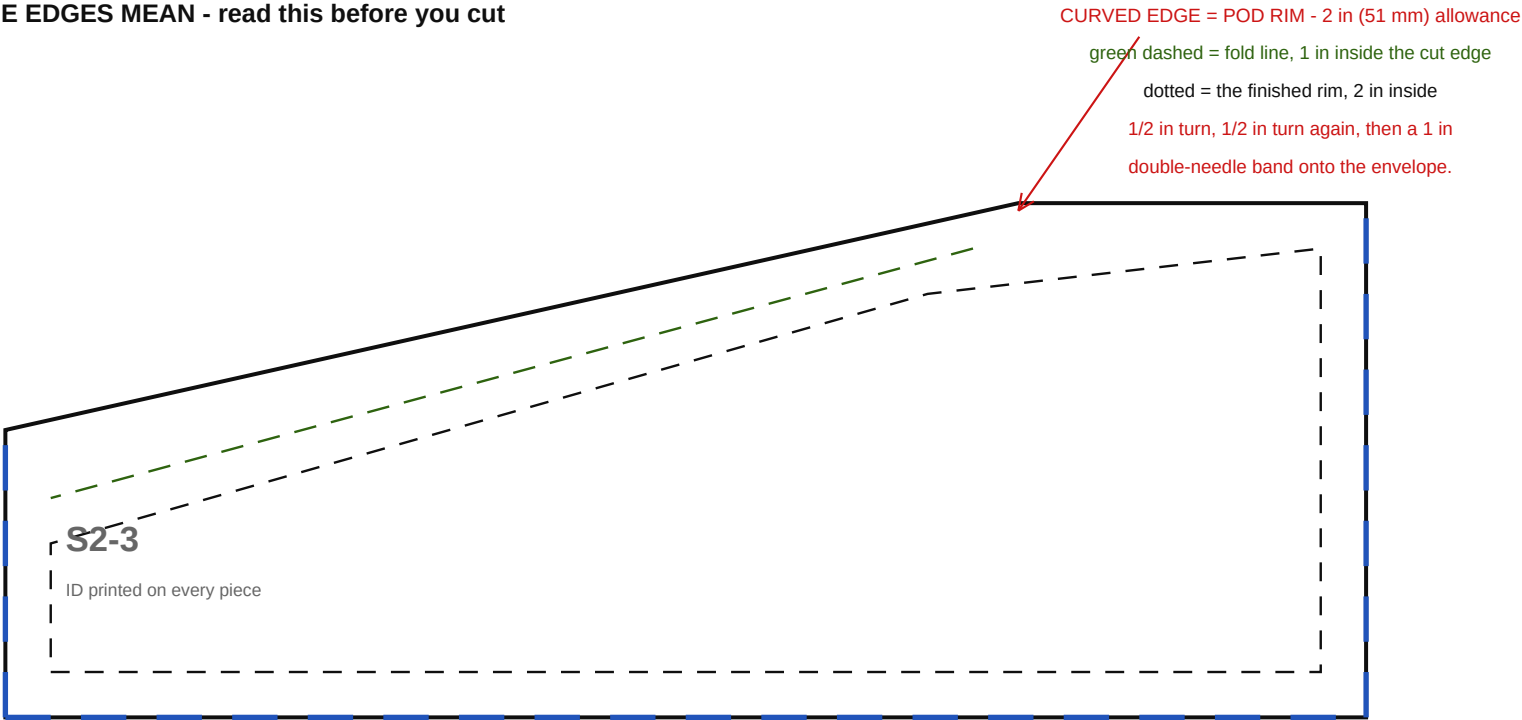
1 PRINT AND CHECK

Print alien-balloon-eye-patterns.pdf at 100 % on a 60 in roll - 14 pages, one per piece. Every page carries calibration lines - measure them in BOTH directions before you cut anything. A 2 % print error over a 3.4 m piece is 68 mm, and it will show up as a rim that does not close.

2 CUT THE PIECES

Hot-knife every edge. Cut each piece TWICE - once as printed, once mirrored. Label every piece on the WRONG side, in the corner, with its ID and L or R, plus an arrow pointing along the strip. The pieces are nearly rectangular and easy to get 180 degrees out.

WHAT THE EDGES MEAN - read this before you cut

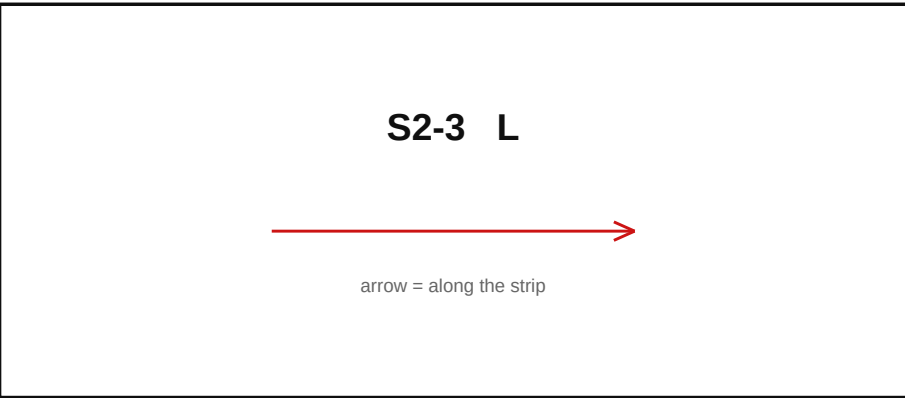


STRAIGHT EDGE = STRIP SEAM. 25 mm, fabric to fabric.
The dotted line 25 mm inside is where you sew.

COMMON MISTAKE: cutting both pods from the same face of the cloth. The two pods are mirror images. If you stack two layers face to face and cut once, you get one L and one R - which is exactly what you want.

Check as you go: lay the cut pieces out in the piecing-plan pattern on sheet 2 before you sew anything.

LABEL EVERY PIECE ON THE WRONG SIDE

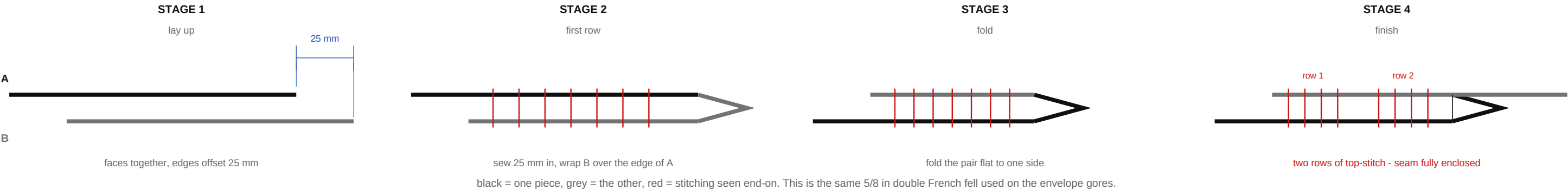


piece ID + L or R + direction arrow
Black on black is hard to read later - use a silver pen.

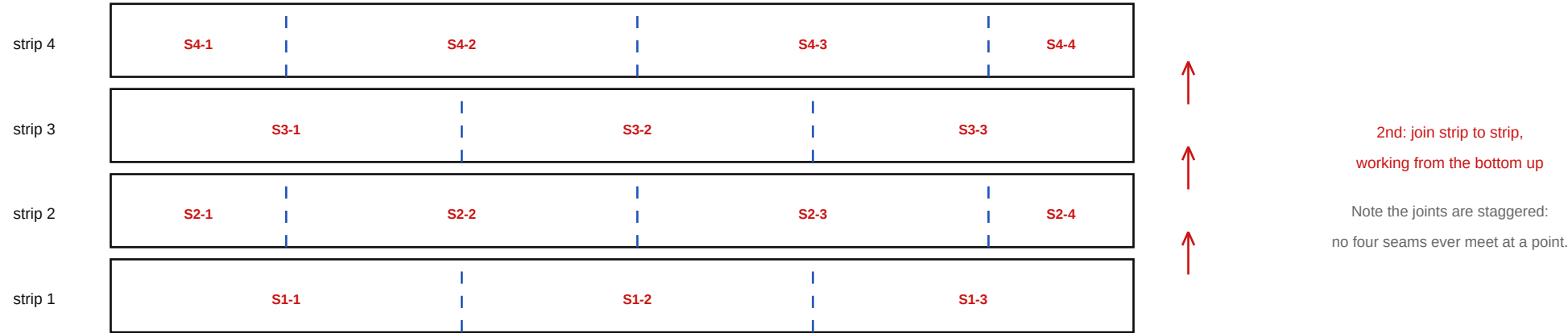
STEP 3 - SEAM THE STRIPS

5/8 in double French fell, the same seam as the envelope gores - sections within a strip first, then strip to strip. This is the whole of the skin's assembly.

THE SEAM, STAGE BY STAGE - cross-section, cloth thickness hugely exaggerated



ORDER OF ASSEMBLY - sections first, then strips



1st: join the sections along each strip - the dashed joints

3a SECTIONS FIRST

Join the sections within each strip. Match the piece IDs printed on the templates - S2-1 to S2-2 to S2-3 and so on. Press each seam as you go.

3b THEN STRIP TO STRIP

Join strip 1 to strip 2, then 2 to 3, then 3 to 4. These seams must be dead straight: they are the datum the whole shape hangs off. Snap a chalk line and sew to it rather than following the cut edge.

3c CHECK THE LENGTH

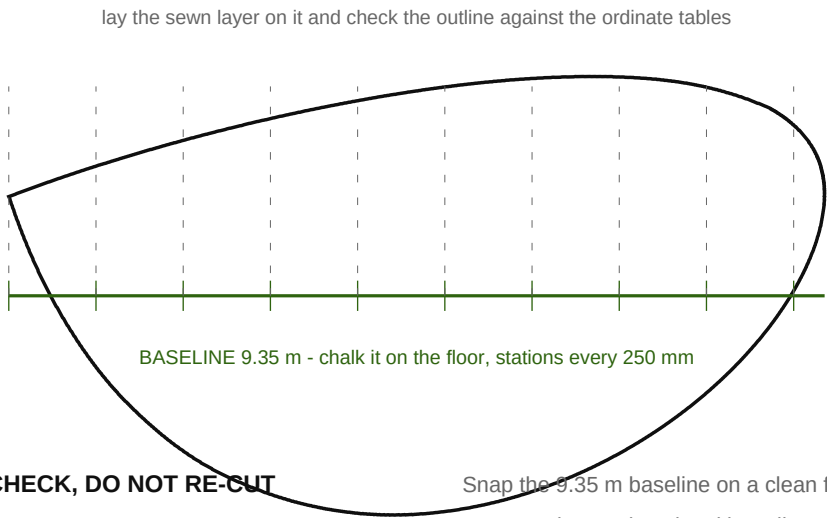
After each strip seam, measure the assembly across the seam. If it is growing, you are taking less than 25 mm somewhere - stop and fix it now, not at the rim.

Thread V-69. Needle size 16-18. All edges hot-cut - a raw ripstop edge inside a fell will still fray in service.

STEPS 4-5 - CHECK THE ASSEMBLED LAYER, THEN MARK THE ENVELOPE

the one place where the full-size loft is still used - as a check, not as a pattern

CHECK AGAINST THE BASELINE LOFT



4 CHECK, DO NOT RE-CUT Snap the 9.35 m baseline on a clean floor. Lay the assembled skin on it and check the outline at every station against the skin ordinates in alien-balloon-eyes.pdf (sheet 5). Tolerance 10 mm. If it is out by more, the error is in a strip seam - unpick and re-sew rather than trimming the rim.

5 MARK THE ENVELOPE With the gores joined and the envelope flat, transfer the footprint outline onto it. The red eye outline is already printed on the F to J panel templates, so this is really just joining marks that are already there. THIS OUTLINE IS THE POD'S BASE - the envelope panels inside it are the other half of the pod. Mark the numbered match points round it at the same time.

WHERE THE OUTLINE COMES FROM



the red outline is printed on panels F-J; join it up across the seams

THERE IS NO BACKING LAYER

Earlier issues of this pack put a full black footprint layer between the skin and the envelope. It was dropped. It had no structural job - the 25 mm rim tape carries the pod load into the gore seams either way - and it cost 34 m2 of cloth and 14 extra pieces per pod, made the rim a four-ply stack, and doubled the feed-hole hemming.

So: mark the outline, tape it, sew the skin to it. Two layers, not three.

Feed holes sit 700 mm or more inside the rim, so the hems never reach the tape.

Do this after the gores are joined and before the parachute is rigged: the envelope must lie flat, and you need to reach both faces to sew the rim tape.

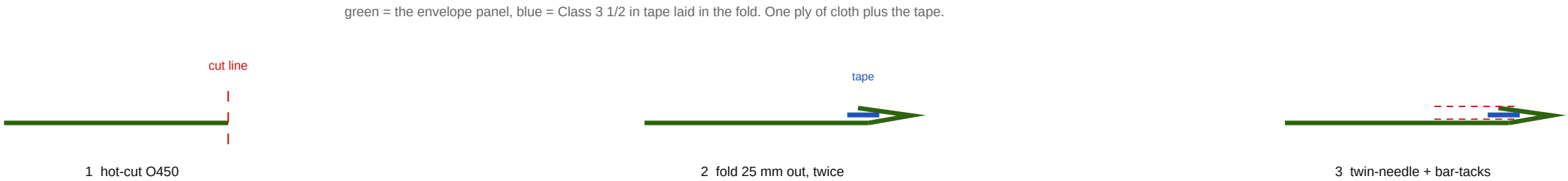
STEP 6 - FEED HOLES

three per pod, cut through the envelope and the footprint layer together - these are what inflate the pod

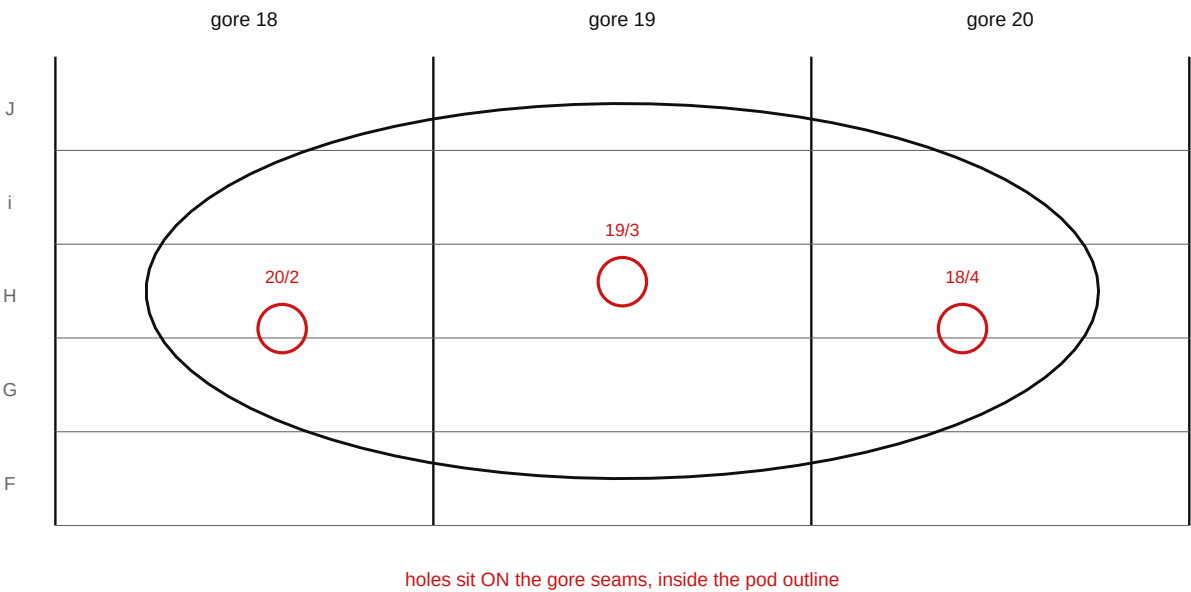
6 CUT AND HEM

Cut these in the ENVELOPE PANELS, before the pod skin goes on. Positions: gores 20/2, panel H, 450 mm up from the row bottom; gores 19/3, panel H, 910 mm up from the row bottom; gores 18/4, panel i, 850 mm up from the row bottom. Hot-cut at O450, then double-fold the edge outward to finish at O500. Lay one layer of Class 3 1/2 in tape in the fold and twin-needle through it. Bar-tack at four points round each hole. One ply of cloth - there is no backing layer, so the tape is what stops the hem tearing out.

HOLE HEM - CROSS-SECTION



WHERE THEY GO



WHY THREE, AND WHY ON THE SEAMS

Three holes spread the feed so the pod fills evenly rather than ballooning at one end. On the seams, the load tape carries the hole loads instead of the panel cloth.

Total open area is about 0.59 m2 per pod - enough to fill the pod at the same rate as the envelope, so it stays taut in flight instead of flapping.

DO NOT cut the holes in the pod skin. The skin is closed; the pod inflates from inside the envelope.

Cut the holes BEFORE you sew the rim - reaching inside a closed pod with a hot knife is not fun.

STEPS 7-8 - RIM TAPE, THEN EASE THE SKIN ON

the outer skin rim is 1.24 m longer than the footprint rim - that surplus is the dome, and it must be spread evenly

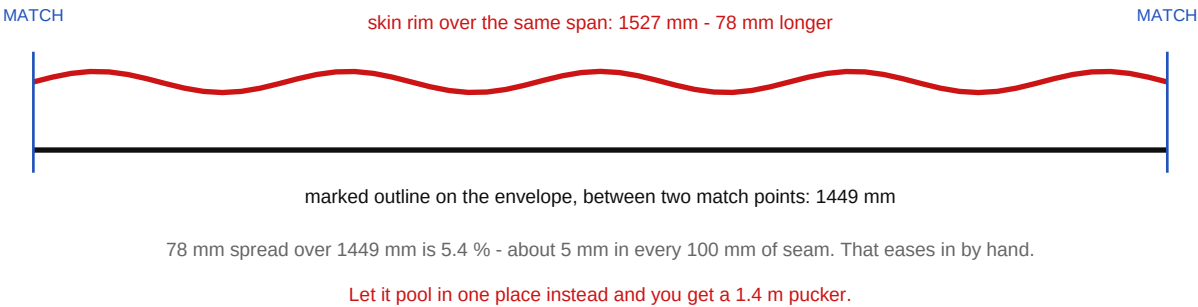
7 RIM TAPE

Sew 23.2 m of Class 3 1/2 in tape round the outline marked on the ENVELOPE, twin-needle, lapping the join 150 mm. The tape, not the panel cloth, carries the pod load - so where the outline crosses a gore seam, catch it into that seam's vertical load tape. That is what replaces the old backing layer.

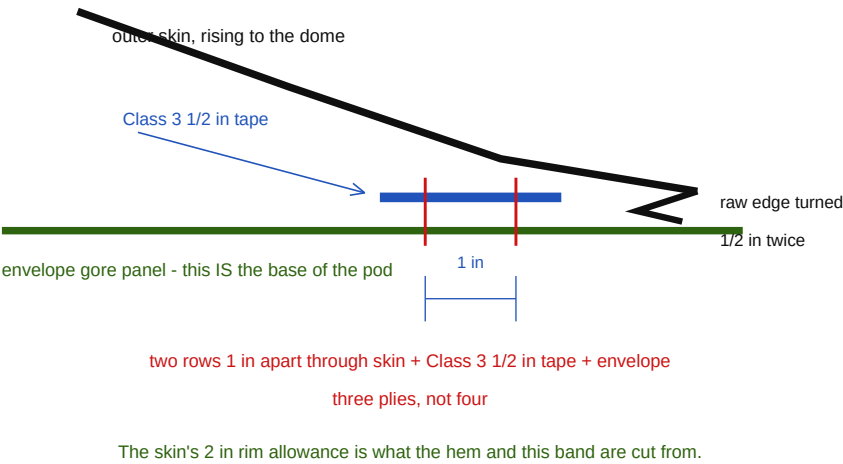
8 MATCH AND EASE

Pin the skin to the taped outline matching numbers 1 to 16. Each segment holds 78 mm of surplus, worked in evenly - little tucks you can smooth out with your thumb, not one big pleat. Turn the raw edge under 1/2 in, then 1/2 in again over the tape, and run TWO rows 1 in apart. Leave one segment open.

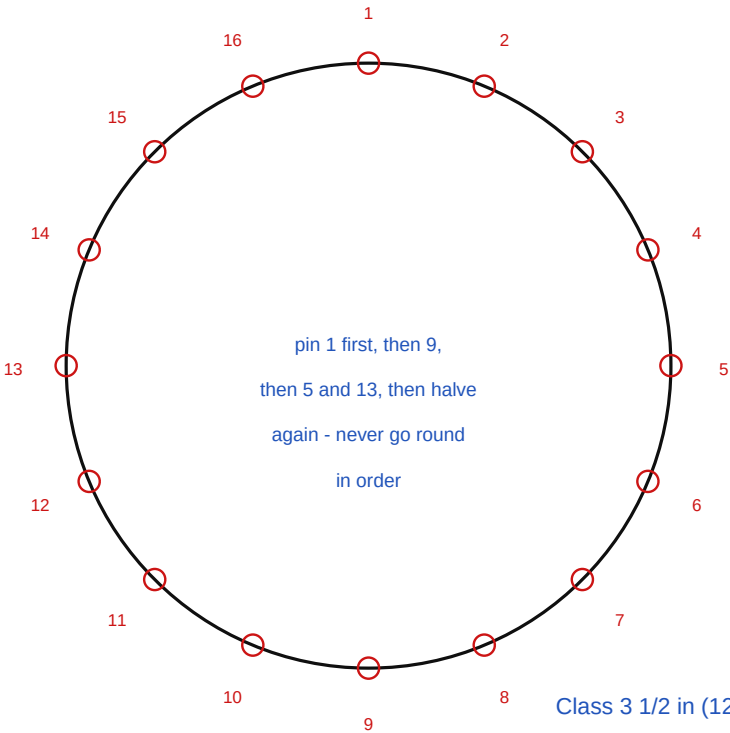
WHAT '78 mm PER SEGMENT' LOOKS LIKE



RIM SEAM - CROSS-SECTION



PINNING ORDER - this is what stops the surplus pooling



Pinning 1, 2, 3, 4 ... in order pushes all the accumulated surplus into the last segment. Halving instead - 1, then the opposite point, then the two quarters, then the eighths - shares the error out and it disappears.

Leave one segment open so you can reach inside and check the feed holes are clear before you close it up.

If a segment will not lie flat, the fault is upstream: a strip seam that took less than 25 mm, or a piece cut from a page that printed short.

Class 3 1/2 in (12.7 mm) load tape: 23.2 m at the rim per pod. Skin rim 24.43 m against marked outline 23.18 m.

STEPS 9-10 - CLOSE, LOAD TEST, INFLATE

the pod carries real load in flight - prove it on the ground

- 9

CLOSE THE LAST SEGMENT

Reach in, check the three feed holes are clear and nothing is trapped inside, then close the last segment the same way as the rest: two rows, 6 mm apart, through skin + tape + envelope.
- 10

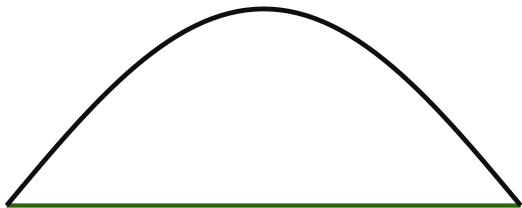
LOAD TEST

Pull 25 kg at four points round the rim - the two ends of the long axis and the two sides. Hold each for 30 seconds. Look for tape pulling out of the stitching, not for the cloth stretching; ripstop stretches and recovers, a failing seam does not.
- 11

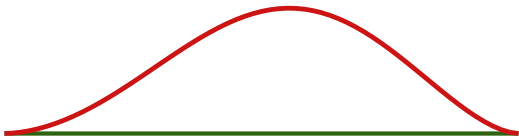
GROUND INFLATION

Cold-inflate the envelope and look at the pod from 20 m. It should stand about 1300 mm proud, evenly full, with no flat spot at either end of the long axis and no standing wrinkle running out from a match point. A soft end means a blocked feed hole; a wrinkle means eased surplus that pooled.

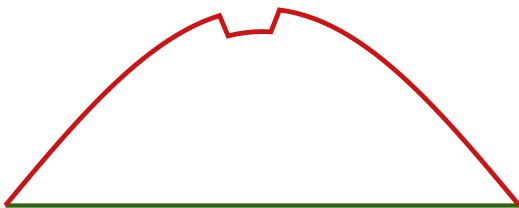
WHAT GOOD LOOKS LIKE, AND WHAT IT DOES NOT



GOOD - even, full, symmetric



SOFT END - blocked or undersized feed hole



STANDING WRINKLE - ease pooled at one match point

SIGN-OFF - copy these onto build log sheet 3

- ☐14 skin pieces cut, L and R, all labelled
- ☐Strip seams sewn and measured - no growth across the seams
- ☐Assembled skin checked against the loft, within 10 mm at every station
- ☐Footprint outline and match points marked on the envelope
- ☐Three feed holes cut and hemmed over tape in the ENVELOPE per pod, O500 finished
- ☐23.2 m of Class 3 1/2 in tape sewn to the outline, join lapped 150 mm, caught into the gore load tapes
- ☐Skin eased on over 16 match points, hem turned twice, two rows 1 in apart
- ☐25 kg load test at four points - passed
- ☐Ground inflation - 1300 mm proud, even, no wrinkles

date

date

date

date

date

date

date

date

date

TROUBLESHOOTING

every one of these has the same root cause: something was measured once instead of twice

The rim will not close - the skin is much too big

You cut the skin from the footprint ordinates by mistake, or vice versa. The skin is 224 mm bigger all round and its pieces are lettered S, not F.

The rim will not close - the skin is slightly too big everywhere

Normal. It IS 5.4 % bigger - that is the dome. Pin the 16 match points first and work the surplus in between them.

Surplus all ends up in the last segment

You pinned in order round the rim. Unpin, pin 1, then the opposite point, then quarters, then eighths.

A strip seam is visibly curved

It was sewn to the cut edge instead of a chalk line. Unpick it - a curved strip seam distorts the whole layer and the rim will never fair up.

The assembly is longer than the ordinates say

A seam took less than 25 mm. Each 1 mm lost per seam over 4 strips is 4 mm of growth; over the sections too it compounds. Re-sew the offending seam.

The assembly is short by a consistent percentage

The templates printed at 'fit to page'. Check the calibration lines on the printed sheets - they must measure exactly what they say. Re-print at 100 %.

Pod inflates soft at one end

A feed hole is blocked, undersized, or was never cut - check all three from inside the envelope. They are cut in the envelope panels only.

Pod is full but flaps in flight

The rim tape is missing, or was sewn to panel cloth without being caught into the gore-seam load tapes. With no backing layer the tape is the only load path - it must be continuous and tied into those seams.

Left and right pods do not match

Both were cut from the same face. They are mirror images - stack two layers face to face and cut once.

Pod pulls a pucker in the envelope when cold

Expected to a small degree when deflated. If it is still there when inflated, the outline was marked smaller than the skin - re-mark from the red outline on the F to J panel templates.

The envelope cloth is stretching round the rim stitching

The rim tape is not carrying the load into the gore seams. With the backing layer gone the tape must be caught into every vertical load tape it crosses; otherwise the panel cloth takes the pull.

Airworthiness: appendages change how an envelope behaves. Have the finished pods inspected by an experienced balloon repair station before the first flight.